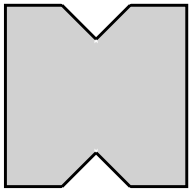
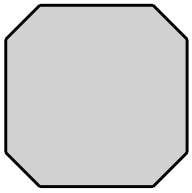
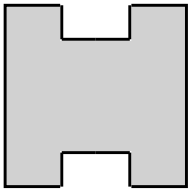
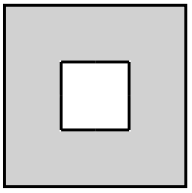
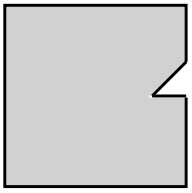
Preview: y2007_q24 (v1)

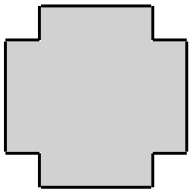
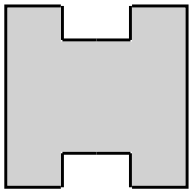
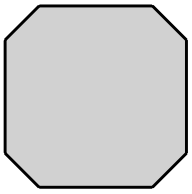
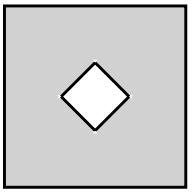
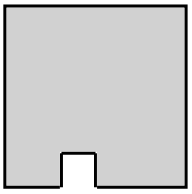
Fold twice, cut, unfold: one cut is mirrored over both folds (4 copies, symmetric)

Part A: Each question is worth 3 points

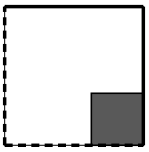
1. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square is cut off, and the paper is unfolded. Which design *cannot* be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

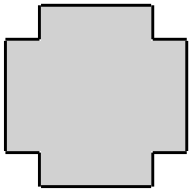
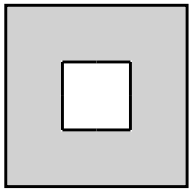
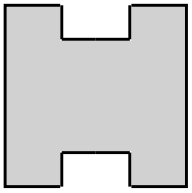
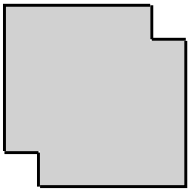
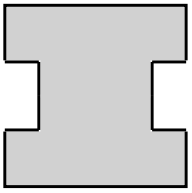
2. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square is cut off, and the paper is unfolded. Which design *cannot* be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

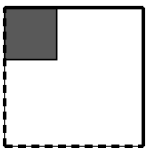
3. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). The dark part is cut off, then the paper is unfolded.



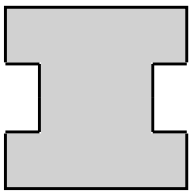
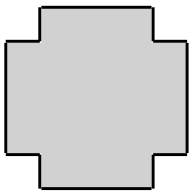
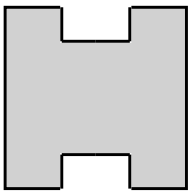
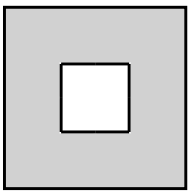
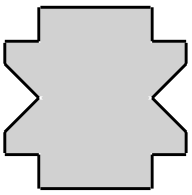
What does the paper look like?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

4. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). The dark part is cut off, then the paper is unfolded.

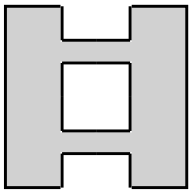
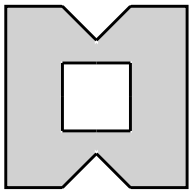
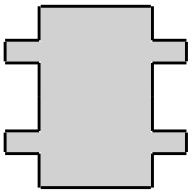
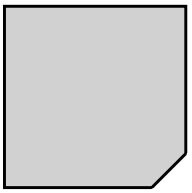
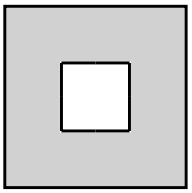


What does the paper look like?

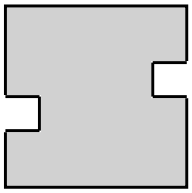
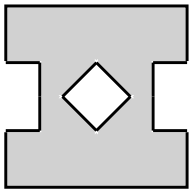
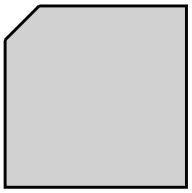
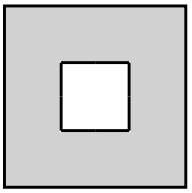
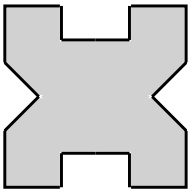
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part B: Each question is worth 4 points

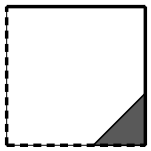
5. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square is cut off, and the paper is unfolded. Which design *can* be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square is cut off, and the paper is unfolded. Which design *can* be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

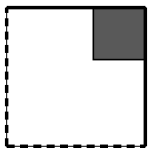
7. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). The dark part is cut off. Before unfolding, it is one small piece, but it is made of several layers.



When the cut-off piece is unfolded too, how many separate pieces of paper is it?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 8

8. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). The dark part is cut off. Before unfolding, it is one small piece, but it is made of several layers.

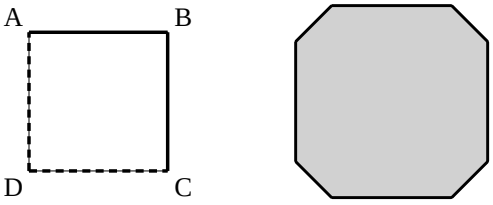


When the cut-off piece is unfolded too, how many separate pieces of paper is it?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 8

Part C: Each question is worth 5 points

9. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square (left picture) was cut off, and after unfolding the paper looks like the right picture.

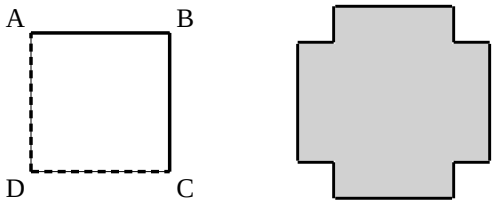


Which corner was cut off?

A. corner A ***B.*** corner B ***C.*** corner C

D. corner D ***E.*** it cannot be made with one cut

10. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square (left picture) was cut off, and after unfolding the paper looks like the right picture.



Which corner was cut off?

A. corner A

B. corner B

C. corner C

D. corner D

E. it cannot be made with one cut

Answer Key

1. **E**	2. **E**	3. **E**	4. **C**	5. **E**	6. **D**
7. **B**	8. **D**	9. **B**	10. **B**		

Solutions

1. E — The cut goes through four layers, so it appears four times, mirrored over both folds. Design E is impossible: it has only one cut part, but the cut goes through four layers. [2007 Q24 · which_cannot]

2. E — The cut goes through four layers, so it appears four times, mirrored over both folds. Design E is impossible: it has only one cut part, but the cut goes through four layers. [2007 Q24 · which_cannot]

3. E — The cut corner is the middle of a side of the paper (look where the folds are). Mirrored four times it gives notches in the middle of the left and right sides: design E. [2007 Q24 · which_result]

4. C — The cut corner is the middle of a side of the paper (look where the folds are). Mirrored four times it gives notches in the middle of the top and bottom sides: design C. [2007 Q24 · which_result]

5. E — A single cut appears four times, the same on the left and right and on the top and bottom. Only design E looks like that (a hole in the middle). [2007 Q24 · which_can]

6. D — A single cut appears four times, the same on the left and right and on the top and bottom. Only design D looks like that (a hole in the middle). [2007 Q24 · which_can]

7. B (2) — The cut is at the middle of a paper side: the layers are joined in pairs along one fold, so the cut gives 2 pieces (two notches). [2007 Q24 · count_pieces]

8. D (4) — The cut is at a real corner of the paper: the four layers are not joined there, so it gives 4 separate pieces (the four corners). [2007 Q24 · count_pieces]

9. B (corner B) — The design shows all four corners cut off. That comes from cutting corner B (the real paper corner, away from the folds). [2007 Q24 · which_corner]

10. B (corner B) — The design shows all four corners cut off. That comes from cutting corner B (the real paper corner, away from the folds). [2007 Q24 · which_corner]